Circles Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Radius vs. diameter

ORIGINAL PROBLEM

Find the area of a circle with diameter 10.

INCORRECT WORKED SOLUTION

$A = \pi r^2$

$A = \pi(10)^2$

$A = 100\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

The formula uses the RADIUS, which is half the diameter: r = 5. $A = \pi(5)^2 = 25\pi$.

B. SHOW CORRECT WORK (1 POINT)

$r = \frac{10}{2} = 5$

$A = \pi(5)^2$

$A = 25\pi$

2. Circumference

ORIGINAL PROBLEM

Find the circumference of a circle with diameter 12.

INCORRECT WORKED SOLUTION

$C = 2\pi r$

$C = 2\pi(12)$

$C = 24\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

The formula $2\pi r$ needs the RADIUS, r = 6. (Or use $C = \pi d = 12\pi$.)

B. SHOW CORRECT WORK (1 POINT)

$r = 6$

$C = 2\pi(6)$

$C = 12\pi$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Arc length

ORIGINAL PROBLEM

Find the length of an arc with central angle 60° in a circle of radius 9.

INCORRECT WORKED SOLUTION

$$\text{Arc} = \frac{60}{360} \cdot \pi r^2$$

$$\text{Arc} = \frac{60}{360} \cdot \pi(81)$$

$$\text{Arc} = \frac{27\pi}{2}$$

A. EXPLAIN THE MISTAKE (1 POINT)

That is the formula for SECTOR AREA. Arc length uses the circumference: $\frac{60}{360} \cdot 2\pi r = \frac{1}{6} \cdot 18\pi = 3\pi$.

B. SHOW CORRECT WORK (1 POINT)

$$\text{Arc} = \frac{60}{360} \cdot 2\pi r$$

$$\text{Arc} = \frac{1}{6} \cdot 2\pi(9)$$

$$\text{Arc} = 3\pi$$

4. Sector area

ORIGINAL PROBLEM

Find the area of a sector with central angle 90° and radius 4.

INCORRECT WORKED SOLUTION

$$\text{Area} = \frac{90}{360} \cdot 2\pi r$$

$$\text{Area} = \frac{1}{4} \cdot 2\pi(4)$$

$$\text{Area} = 2\pi$$

A. EXPLAIN THE MISTAKE (1 POINT)

That is ARC LENGTH. Sector AREA uses πr^2: $\frac{1}{4} \cdot \pi(16) = 4\pi$.

B. SHOW CORRECT WORK (1 POINT)

$$\text{Area} = \frac{90}{360} \cdot \pi r^2$$

$$\text{Area} = \frac{1}{4} \cdot \pi(16)$$

$$\text{Area} = 4\pi$$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Inscribed angle

ORIGINAL PROBLEM

An inscribed angle intercepts an arc of 80°. Find the measure of the inscribed angle.

INCORRECT WORKED SOLUTION

Inscribed angle = arc

Inscribed angle = 80°

A. EXPLAIN THE MISTAKE (1 POINT)

An inscribed angle is HALF of its intercepted arc: $\frac{80}{2} = 40°$. (A CENTRAL angle equals its arc.)

B. SHOW CORRECT WORK (1 POINT)

Inscribed angle = $\frac{1}{2} \cdot$ arc

Inscribed angle = $\frac{80}{2}$

Inscribed angle = 40°

6. Angle inside a circle

ORIGINAL PROBLEM

Two chords intersect inside a circle. The arcs intercepted by the angle and by its vertical angle measure 50° and 90°. Find the angle.

INCORRECT WORKED SOLUTION

Angle = $\frac{90 - 50}{2}$

Angle = 20°

A. EXPLAIN THE MISTAKE (1 POINT)

When chords meet INSIDE a circle, the angle is half the SUM of the intercepted arcs: $\frac{50 + 90}{2} = 70°$. (The difference is for lines meeting OUTSIDE.)

B. SHOW CORRECT WORK (1 POINT)

Angle = $\frac{50 + 90}{2}$

Angle = $\frac{140}{2}$

Angle = 70°

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Tangent lines

ORIGINAL PROBLEM

Tangent segment PT touches a circle at T. The radius OT = 5 and OP = 13. Find PT.

INCORRECT WORKED SOLUTION

PT = OP − OT

PT = 13 − 5 = 8

A. EXPLAIN THE MISTAKE (1 POINT)

A radius is PERPENDICULAR to a tangent at the point of tangency, so △OTP is a right triangle with hypotenuse OP: $PT^2 + 5^2 = 13^2$, so PT = 12.

B. SHOW CORRECT WORK (1 POINT)

OT ⊥ PT (radius ⊥ tangent)

$PT^2 + 5^2 = 13^2$

$PT^2 = 144$

PT = 12

8. Intersecting chords

ORIGINAL PROBLEM

Two chords intersect inside a circle. One is cut into segments 4 and x. The other is cut into 6 and 8. Find x.

INCORRECT WORKED SOLUTION

4 + x = 6 + 8

x = 10

A. EXPLAIN THE MISTAKE (1 POINT)

The Intersecting Chords Theorem MULTIPLIES the segments: 4 · x = 6 · 8.

B. SHOW CORRECT WORK (1 POINT)

4 · x = 6 · 8

4x = 48

x = 12

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Two tangents

ORIGINAL PROBLEM

Two tangent segments from the same point P have lengths PA = 3x + 2 and PB = 5x − 10. Find x.

INCORRECT WORKED SOLUTION

PA + PB = 90

3x + 2 + 5x − 10 = 90

8x = 98

x = 12.25

A. EXPLAIN THE MISTAKE (1 POINT)

Tangent segments from the same external point are EQUAL: 3x + 2 = 5x − 10.

B. SHOW CORRECT WORK (1 POINT)

PA = PB

3x + 2 = 5x − 10

12 = 2x

x = 6, so PA = PB = 20

10. Angle in a semicircle

ORIGINAL PROBLEM

AB is a diameter and C is a point on the circle. Find m∠ACB.

INCORRECT WORKED SOLUTION

Arc AB is a semicircle: 180°.

m∠ACB = 180°

A. EXPLAIN THE MISTAKE (1 POINT)

An inscribed angle is HALF its intercepted arc: $\frac{180}{2} = 90°$.

Every angle inscribed in a semicircle is a right angle.

B. SHOW CORRECT WORK (1 POINT)

Arc AB = 180° (semicircle)

$m\angle ACB = \frac{1}{2} \cdot 180°$

m∠ACB = 90°

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points